present *pMapper*, an application placement controller that dynamically places applications to minimize power while meeting performance guarantees.

System management costs have escalated rapidly with the growing number of densely packed under-utilized machines in the data center. Virtualization is seen as a solution that can provide the required isolation layer to consolidate applications running on a large number of low utilization servers to a smaller number of highly utilized servers. The virtualization layer typically provides flexible runtime mechanisms for fine grain resource allocation. In fact, high speed live migration of virtual machines is also possible between the physical servers in a cluster. This enables applications in virtual machine containers to be moved at runtime in response to changing workload to dynamically optimize the application placement on physical servers. Thus mechanisms to allow dynamic resizing and migration of virtual machine containers among physical servers enables research in dynamic application placement middleware beyond static provisioning.

A second trend that is important for dynamic application placement is the growing awareness about energy consumption in data centers and the significant adverse impact on the environment in terms of CO_2 emissions from the cooling systems. The current power density of data centers is typically around 100 Watt per sq.ft. and growing at the rate of $15 - 20\%$ per year [17]. Thus, it is increasingly being realized that inefficient use of servers in a data center leads to high energy costs, expensive cooling hardware, floor space, and also adverse impact on the environment. There is a large initiative in the industry as well as academia to develop technologies that will help to create "green" or environment-friendly data centers that will optimize energy consumption and consolidate hardware resources, besides being sensitive to application performance and availability SLAs. To this extent, a dynamic application placement controller can use virtualization to resize VM containers of applications or migrate VMs at runtime to consolidate the workload on an optimal set of physical servers. Servers unused over a period of time can be switched to low power states to save energy. Further, we observed a fairly large dynamic power range (e.g., the static power of an IBM HS-21 blade was 140Watt and the dynamic potentially non-linear power range was almost 80Watt for the `daxpy` benchmark). Hence, even in scenarios where servers can not be switched off by consolidation, power savings can be obtained by packing the servers at optimal target utilization.

The use of power management techniques during application placement has its own implications. A dynamic application placement controller uses live migration and we found from our testbed experimentation that the cost of live migration is significant, and needs to be factored by the dynamic placement controller. For example, a 512MB VM running HPC benchmarks require almost a minute to migrate and causes a 20-25% drop in application throughput during the live migration. We also observed that a large number of live migrations are required for dynamic placement, thus emphasizing the importance of taking migration cost into account.